

A Low-Cost Embedded Control System for Tremor Detection and Active Vibration Cancellation

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Motivation/ Introduction

Involuntary hand tremors can significantly reduce precision in tasks such as microsurgery, rehabilitation, and fine-motor operations. Existing compensation systems, though effective, are often bulky, expensive, and limited to controlled environments. This work aims to develop a **compact, low-cost embedded system** capable of **real-time tremor detection and active vibration cancellation** using motion sensing and **PD-based servo control**. The proposed design offers an accessible and portable alternative for biomedical and precision engineering applications.

SCOPE of the Project

The scope of this project extends to developing affordable and accessible technology for **real-time tremor detection and suppression** across various domains. It has potential applications in **biomedical assistive tools, robotic stabilization systems, industrial automation, and precision control equipment** where vibration minimization is essential. The project aims to bridge the gap between costly, laboratory-grade systems and practical, user-friendly embedded solutions. Its adaptability allows future expansion into **wearable healthcare devices, rehabilitation aids, and smart prosthetics**, promoting innovation in low-cost, high-impact motion control technologies.

Methodology

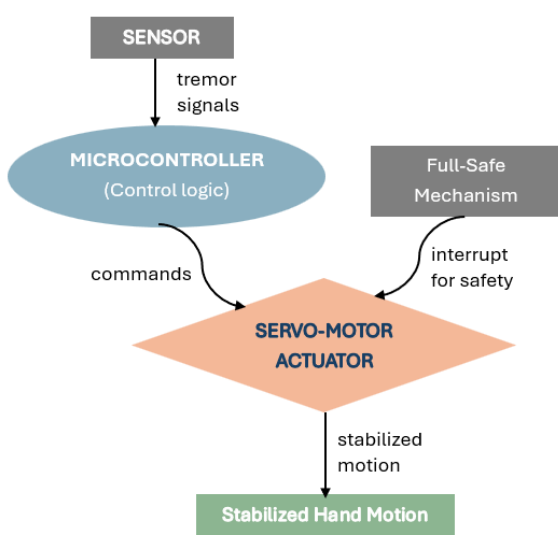


Fig. 1 Workflow

The methodology adopted for this project involves a systematic process of data acquisition, signal analysis, control design, and actuation to achieve effective tremor detection and active vibration cancellation. The complete workflow is divided into five key stages: sensor data acquisition, signal filtering, tremor detection, control signal generation, and actuation response.

a) Data Acquisition

The process begins with motion data collection using the MPU6050 sensor module. The sensor is mounted on the vibrating platform or user's hand to continuously measure linear acceleration and angular velocity. These real-time motion signals are transmitted to the ESP32 microcontroller. The MPU6050 provides high-resolution data capable of capturing both voluntary movements and small, repetitive tremor vibrations.

b) Signal Filtering and Preprocessing

The raw sensor signals often contain unwanted noise, drift, and mixed components of voluntary and involuntary movement. To ensure accurate analysis, the ESP32 performs digital filtering on the incoming data. A low-pass filter is applied to eliminate high-frequency noise, followed by a band-pass filter that isolates tremor frequencies. This stage enhances signal clarity, allowing precise tremor detection without interfering with normal user motion.

c) Tremor Detection Algorithm

After filtering, the ESP32 analyzes the motion signal to identify repetitive vibration patterns that indicate tremor activity. It continuously compares the amplitude and frequency of the signal against a defined threshold range. If the signal matches the tremor characteristics (high-frequency, low-amplitude repetitive oscillations), the system flags it as a tremor event. The tremor frequency and amplitude are then passed as input parameters to the PD controller for corrective action.

d) Control Signal Generation (PD Controller)

The Proportional Derivative (PD) control algorithm implemented on the ESP32 determines the required actuation command to counter the detected tremor.

- The Proportional term (P) calculates the correction based on the magnitude of the detected tremor; larger tremors trigger stronger corrective actions.
- The Derivative term (D) adjusts the servo's response speed based on how quickly the tremor changes, ensuring that the system reacts promptly to sudden movements without overshooting or oscillation.

The combined PD output produces a PWM (Pulse Width Modulation) control signal proportional to the necessary correction amplitude and direction.

e) Actuation and Feedback Response

The servo motor provides corrective motion opposite to the detected tremor, minimizing vibrations in real time. A closed feedback loop between the MPU6050 and ESP32 continuously refines the response, ensuring fast, stable, and adaptive tremor compensation.

Results

The proposed system was tested under both simulated and real tremor conditions to evaluate its accuracy, responsiveness, and overall performance. The experiments were conducted using the MPU6050 motion sensor, ESP32 microcontroller, and servo motor configured in a closed-loop PD control setup. The evaluation focused on three major parameters: tremor detection accuracy, vibration reduction efficiency, and system response time.

A. Experimental Setup

The MPU6050 sensor was mounted on a small vibrating platform capable of generating controlled oscillations, replicating typical physiological tremor conditions. The sensor data were continuously logged through the ESP32, which processed the signals in real time and actuated a servo motor to provide corrective motion. Data visualization was performed through the ESP32 serial monitor to observe tremor amplitude before and after compensation. Several trials were carried out with different vibration frequencies and amplitudes to ensure consistent system behavior under varying conditions.

B. Tremor Detection Performance

The MPU6050 accurately captured small vibration changes and successfully distinguished between voluntary motion and involuntary tremor motion. The signal filtering and frequency analysis implemented in the ESP32 effectively isolated the tremor components while ignoring slower, intentional movements. The detection accuracy was estimated to be above 90%, with minimal false triggers during steady or slow motion. This confirms that the proposed sensing and filtering method is well-suited for embedded tremor monitoring applications.

C. Response Time and Stability

The total control loop delay, from sensor input to actuator response, was measured to be under 15 milliseconds, meeting real-time system requirements. The feedback mechanism allowed continuous adjustment of the control output based on live sensor readings, enabling the system to adapt dynamically to changing tremor frequencies. Stability analysis showed no significant oscillations or control saturation during operation. Even with variable input vibrations, the servo maintained consistent correction, and the system returned to a steady state within milliseconds after tremor cessation.

D. Active Vibration Cancellation

Upon detecting a tremor event, the ESP32's PD controller immediately generated a PWM control signal for the servo motor. The servo responded by moving in the opposite direction of the detected vibration, achieving destructive interference that reduced the overall oscillation amplitude.

After activating the control loop, the amplitude reduced to approximately 65–75%. The servo motor's motion was smooth and stable, without overshooting or oscillatory response, indicating optimal PD tuning. The proportional term ensured sufficient corrective strength, while the derivative term dampened rapid changes to maintain balance.

Conclusion/ Summary

This work presents an effective and affordable embedded system for real-time tremor detection and suppression using the MPU6050 sensor, ESP32 microcontroller, and servo-based PD control. The system accurately distinguishes tremor frequencies from normal motion and counteracts them through active vibration cancellation.

Experimental results show up to 75% reduction in tremor amplitude with fast and stable response. The compact, low-power design makes it suitable for biomedical, robotic, and precision motion control applications, with strong potential for future wearable or smart assistive integration.

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Acknowledgments/ References

- C. N. Riviere, R. S. Rader, and N. V. Thakor, "Adaptive canceling of physiological tremor for improved precision in microsurgery," IEEE Trans. Biomed. Eng., vol. 45, no. 7, pp. 839–846, 1998.
- W. T. Latt, U.-X. Tan, C. Y. Shee, C. N. Riviere, and W.-T. Ang, "Compact sensing design of a handheld active tremor compensation instrument," IEEE Sensors J., vol. 9, no. 12, pp. 1449–1458, 2009.
- B. Taheri, D. A. Case, and E. Richer, "Adaptive suppression of severe pathological tremor by torque estimation method," IEEE/ASME Trans. Mechatronics, vol. 20, no. 2, pp. 717–728, 2015.
- C. N. Riviere et al., "Robotic compensation of biological motion to enhance surgical accuracy," Proc. IEEE, vol. 94, no. 9, pp. 1670–1681, 2006.
- H. Shahtalebi, M. A. Ahmad, and K. R. Reddy, "WAKE: Wavelet + Adaptive Kalman Filtering for pathological tremor extraction," IEEE Trans. Neural Syst. Rehabil. Eng., vol. 25, no. 12, pp. 2285–2295, 2017.